

TROMSO AP, Norway

WMO: 010250

Lat: 69.683N

Lon: 18.919E

Elev: 9

StdP: 101.21

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.5	-11.1	-16.2	0.9	-10.5	-14.8	1.0	-9.8	15.5	1.9	13.9	0.7	1.9	120	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	20.3	14.5	18.3	13.5	16.5	12.7	15.2	19.2	14.0	17.5	13.1	15.9	3.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	9.5	16.6	12.2	8.9	15.3	11.4	8.4	14.4	42.5	19.1	39.5	17.6	37.0	15.9	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.1	11.4	10.1		DB	-14.6	23.6	1.5	2.4	-15.7	25.3	-16.6	26.8	-17.5	28.1	-18.6	29.8
(5)				WB	-15.2	17.1	1.5	1.8	-16.3	18.4	-17.2	19.4	-18.0	20.4	-19.1	21.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-3.1	-3.5	-2.0	1.0	5.5	9.2	12.2	11.2	7.9	3.3	0.0	-1.7
	DBStd	6.42	4.04	4.10	3.54	3.11	3.20	2.93	2.85	2.54	2.87	3.45	3.46	4.00
	HDD10.0	2582	407	377	373	269	144	49	9	14	73	208	298	362
	HDD18.3	5453	665	610	631	518	397	274	192	220	312	466	548	620
	CDD10.0	171	0	0	0	0	5	25	76	53	12	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	4.3	5.2	5.2	5.0	4.0	4.0	4.1	3.6	3.4	3.5	4.1	4.7	5.2
(15) Precipitation	PrecAvg	1002	99	89	81	61	51	55	67	75	100	131	97	108
	PrecMax	1596	340	248	182	185	147	163	162	192	233	334	242	265
	PrecMin	643	1	2	8	5	4	6	4	16	6	18	9	2
	PrecStd	203	61	49	45	35	30	33	34	42	53	63	50	57
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	5.6	6.3	11.3	18.3	21.4	23.5	22.1	17.0	12.8	9.1	7.2
		MCWB	3.8	3.3	3.1	6.8	11.5	14.3	16.8	15.5	12.5	9.4	7.1	5.0
	2%	DB	4.8	4.4	5.0	8.8	14.7	18.2	21.0	19.2	15.0	10.7	7.7	5.9
		MCWB	3.1	2.5	2.8	5.4	9.2	12.4	15.3	14.1	11.8	8.2	5.8	4.3
	5%	DB	3.7	3.3	4.0	6.7	12.4	15.9	19.0	17.0	13.4	9.5	6.2	4.9
		MCWB	2.2	1.8	2.0	4.2	7.9	11.4	14.2	13.3	10.6	7.5	4.3	3.1
	10%	DB	2.6	2.2	3.0	5.4	10.4	13.8	17.0	15.3	12.1	8.2	5.1	3.8
		MCWB	1.2	0.8	1.2	3.2	6.9	10.3	13.3	12.4	9.9	6.6	3.4	2.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.8	3.8	7.2	12.0	14.7	17.5	16.5	13.4	9.9	7.5	5.5
		MCDB	5.1	5.0	5.1	10.0	18.0	20.2	22.6	20.3	16.0	11.8	8.7	6.3
	2%	WB	3.2	2.8	3.1	5.7	9.7	13.0	15.7	14.7	12.0	8.7	6.0	4.5
		MCDB	4.2	3.9	4.5	8.0	14.2	17.1	20.0	18.0	14.1	10.3	7.2	5.7
	5%	WB	2.3	2.0	2.2	4.5	8.3	11.7	14.6	13.6	11.1	7.7	4.6	3.2
		MCDB	3.5	3.2	3.7	6.4	11.6	15.1	18.4	16.5	13.1	9.1	6.0	4.5
	10%	WB	1.2	0.9	1.3	3.4	7.2	10.6	13.5	12.7	10.2	6.8	3.4	2.1
		MCDB	2.4	2.0	2.8	5.1	9.9	13.4	16.6	15.1	11.9	8.1	4.8	3.5
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.6	4.8	5.2	5.4	5.2	5.7	5.6	5.2	4.3	4.1	4.4
		MCDBR	4.3	4.4	4.1	6.4	9.1	8.8	9.7	9.3	7.4	4.8	4.8	4.5
		MCWBR	3.7	3.7	3.2	4.2	4.9	4.6	5.1	5.1	4.5	3.7	3.9	3.8
	5% WB	MCDBR	4.2	4.4	3.9	5.8	8.4	8.1	9.0	8.2	6.8	4.5	4.5	4.5
		MCWBR	3.9	4.0	3.3	4.0	4.8	4.6	5.0	4.9	4.4	3.8	4.0	4.0
(40) Clear-Sky Solar Irradiance	taub	N/A	0.276	0.324	0.341	0.352	0.350	0.354	0.342	0.322	0.283	N/A	N/A	N/A
	taud	N/A	2.046	2.098	2.210	2.365	2.452	2.476	2.495	2.499	2.192	N/A	N/A	N/A
	Ebn at Noon	N/A	427	656	785	829	845	827	794	704	425	0	0	0
	Edh at Noon	N/A	43	82	100	98	93	88	77	58	36	0	0	0
(44) All-Sky Solar Radiation	RadAvg	0.01	0.26	1.34	3.17	4.58	4.81	4.35	3.09	1.63	0.50	0.00	0.00	0.00
	RadStd	0.00	0.05	0.18	0.31	0.30	0.61	0.45	0.25	0.19	0.08	0.00	0.00	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.60	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page